Pharma TRP

It’s a proposal for building a Pharmaceutical Supplier Audit Management System using ASP.NET Core + Angular + SQL Server.
The system is meant to digitize, track, and manage supplier audits, including official FDA forms and CAPA (Corrective and Preventive Action) tracking.

1. Executive Summary

Problem: Pharma supplier audits are currently manual → slow, inconsistent, risky.
Solution: Build a digital system to handle supplier audits + FDA forms + CAPA.
Benefits: Faster audits, better compliance, reduced risk.

2. Project Scope

Supplier Audit Management Module
Features include:
Audit planning & scheduling
Integration of FDA forms: 482, 483, 484, 463
Findings & CAPA tracking
Real-time dashboards & reports
Regulatory compliance tracking (FDA, EMA, WHO)

3. Technology Stack

Backend → ASP.NET Core (C#)
Frontend → Angular
Database → MS SQL Server
Hosting → Windows Server + IIS

4. Methodology

Agile (Scrum) development → work in sprints (1–2 weeks).
Regular demos, testing, and client feedback.
Tools: GitHub, Jenkins (CI/CD), Selenium (Testing).

5. Deliverables & Timeline

Multiple Sprints (phased delivery):
Analysis & Design
Supplier Audit Module
Quality Management
Facilities Management
Document Control
Supply Chain Management
Regulatory Compliance

Material Management
Manufacturing Execution
Deployment & Documentation

6. Budget & Payments

Work is estimated in person-hours.
Payment linked to milestones (advance, delivery stages).
Change requests = extra cost/time.

7. Benefits for Client

Stronger compliance with FDA rules.
Faster audit closures.
Better risk management & transparency.
Scalable system (future-proof).

This proposal suggests building a digital Pharma Audit Management System (ASP.NET Core + Angular) that will manage supplier audits, FDA(Food and Drug Adminstration) forms, and CAPA(Corrective & Preventive Action) tracking in an Agile way, delivered in multiple sprints.
To understand the client requirement from the document you shared, you should focus on the main goals and expectations highlighted. I'll simplify the key points for you:

1. Client's Core Requirement

The client wants a Pharma Supplier Risk Management system that ensures FDA compliance during inspections, audits, and everyday operations.

2. What They Expect You to Understand

FDA basics:

The FDA regulates pharma products to ensure safety and quality.
Inspections happen in different types (routine, for-cause, pre-approval, directed).

FDA Forms you must handle:

Form 482 – Notice of Inspection.
Form 483 – List of problems/observations.
Form 484 – Samples taken during inspection.
Form 463 – Legal affidavit/confirmation.
CAPA (Corrective and Preventive Action)

Process:

Review issues.
Investigate root causes.
Make a corrective plan.
Implement actions.
Verify effectiveness.
Maintain documentation.

Compliance Workflow:

Prepare (internal audits, training, SOPs).
Receive inspection notice (Form 482).
Allow audit (records, facilities, samples).
Get findings (Form 483) and fix them.
Provide evidence and affidavits (Form 463).
Checklist to Maintain:
Proper SOPs, training records, equipment logs.
Facility cleanliness, segregation, calibration.
CAPA system, supplier qualification, change control.
Testing, OOS (out-of-specification) reports.
Personnel qualifications and FDA readiness.

3. In Simple Words

The client needs a system/process that:
Keeps pharma suppliers and manufacturers ready for FDA inspections.
Ensures all records, SOPs, training, and equipment logs are maintained.
Tracks and responds to FDA forms (482, 483, 484, 463).
Manages CAPA to fix and prevent issues.
Provides an audit trail for compliance.

1. Problem / Issue (Current Situation)

Supplier audits are done manually → slow, inconsistent, error-prone.
Regulatory documentation (FDA Forms 482, 483, 484, 463) is fragmented, sometimes outside compliance systems.
Delayed CAPA tracking (Corrective and Preventive Actions), which increases compliance risks.
No real-time dashboard → management can't see supplier compliance status quickly.
Risk of penalties, audit failure, and product quality issues.

2. Requirement (What Client Needs)

A centralized system for supplier risk and compliance.
Automated handling of FDA forms in the audit process.
End-to-end audit management (planning, execution, reporting).

CAPA management → assign, track, and close issues faster.
Supplier scoring, performance dashboards, and compliance reports.
Integration with ERP/QMS systems and 21 CFR Part 11 compliance.
Real-time dashboards for compliance and risk monitoring.

3. Workflow (Proposed Process)

Audit Planning → Calendar-based scheduling, auditor assignment.
Inspection Notification → Issue FDA Form 482 digitally to supplier.
Audit Execution → Conduct inspection (onsite/remote), record findings.
Sample Collection → If required, issue and log FDA Form 484.
Observations → Record non-conformances in FDA Form 483.
CAPA Management → Assign corrective actions, monitor closure with alerts.
Affidavits → If needed, issue FDA Form 463 for legal statements.
Reporting & Dashboard → Real-time compliance, supplier risk scoring.
Integration → Sync with QMS/ERP for end-to-end compliance tracking.

4. Solution (How System Fixes Problems)

Digitizes and automates the entire audit process.
Embeds FDA forms directly in workflow (482, 483, 484, 463).
Centralized CAPA tracking with reminders/escalations.
Dashboards & Analytics → instant visibility of supplier risk and compliance.
Ensures traceability & data integrity (21 CFR Part 11 compliant).
Improves audit closure time and reduces compliance risks.
Provides supplier scoring for better risk-based decisions.

Problem: Manual, fragmented, high-risk process.
Requirement: Digital, centralized compliance system.
Workflow: Plan → Inspect → Record → Correct → Report → Integrate.
Solution: Automated platform with FDA form integration, CAPA, dashboards.